

M. OMAR NAWAZ

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RESEARCH OVERVIEW

In my interdisciplinary research program, I integrate climate science, air quality modelling, and public health research to inform environmental policymaking. I explore how climate mitigation and air pollution control may improve public health and address Global North/South inequalities using chemical transport modeling, remote sensing, risk assessment, and inverse methods.

PROFESSIONAL APPOINTMENTS

Cardiff University, Cardiff, Wales, UK

2025- Lecturer of Climate Change Science, School of Earth and Environmental Science

George Washington University, Washington, DC, USA

2024 Professorial Lecturer, Milken Institute School of Public Health

2023-2024 NRDC Health Science Policy Fellow, Milken Institute School of Public Health

2023-2025 Postdoctoral Research Associate, Milken Institute School of Public Health

University of Colorado Boulder, Boulder, CO, USA

2018-2023 PhD Research Assistant, Department of Mechanical Engineering

University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

2016-2018 MSc Research Assistant, Gillings School of Global Public Health

EDUCATION

2023 **PhD**, Air Quality Focus, Mechanical Engineering, CU Boulder, Boulder, CO, USA
Research Supervisor: Professor Daven K. Henze

Thesis: An adjoint sensitivity framework for public health: the sources of air pollution and their current and future impacts at the urban and national scale

2018 **MSc**, Environmental Engineering, UNC Chapel Hill, Chapel Hill, NC, USA

Research Supervisor: Professor J. Jason West

Thesis: Benefits of reduced premature mortality from decreases in PM_{2.5} and ozone in the United States from 1999 to 2015

2017 **BSc**, Physics, UNC Chapel Hill, Chapel Hill, NC, USA

2017 **BSc**, Applied Mathematics, UNC Chapel Hill, Chapel Hill, NC, USA

RESEARCH SUPPORT

Total Funded: £636,000; **Total Pending:** £2,110,000

Current

2026 *Health benefits of next generation transportation emissions in Colombia, ICCT, (£2,000), **Consultant***

2026 *Forward and adjoint modeling of transboundary secondary PM_{2.5} in response to climate mitigation, UKRI, (75,000 node-hrs), **PI***

2025-2026 *Maximizing health benefits from reducing oil and gas emissions, Wellcome Trust, (£496,000*), **Institutional PI***

- 2025-2026 *Estimating UK surface-level pollution from satellite data using machine learning and deterministic modeling*, Supercomputing Wales, (100,000 CPU-hrs), **PI**
- 2024-2027 *Application of satellite observations in estimating NO₂ concentrations, mortality burdens, and inequities*, NASA, (£112,000*), **Collaborator**

Pending

- 2026-2032 *ACCEPT: Assessing Climate Change effects on Equity and Public health from Transboundary air pollution*, Wellcome Trust (£2,110,000), **PI**

Past

- 2025-2026 *Health benefits of transitioning to zero emission vehicles by 2050*, International Council on Clean Transportation, (£12,000), **Consultant**
- 2024 *Study of Global Maritime Shipping-Attributable Health Impacts and Policy Benefits*, ICCT, (£5,000), **Consultant**
- 2023-2025 *Study of Global Transportation-Attributable Health Impacts and Policy Benefits*, ICCT, (£9,000*), **Consultant**

Competitive Grants Not Selected for Funding

- 2024 *HEAD-IN: Assessing disaster risk and resilience action benefits associated with compound heat and air quality hazards, exposures, and vulnerabilities*, NASA, (£759,000*), **PI**
- 2020 *Development of a source attribution and data assimilation framework for MAIA primary and secondary target areas in North America and South America*, NASA, (£112,000*), **Future Investigator**

*Funding total converted to GBP from other currency and rounded to the nearest thousand

PUBLICATIONS

Total citations: 788; h-index: 12 (Google Scholar)

Peer-Reviewed Publications (25)

- 2026 Huber, D.E., Kerr, G.H., **Nawaz, M.O.**, Runkel, S., Anenberg, S.C., Goldberg, D.L. TROPOMI NO₂ trends for urban and polluted areas globally from 2019 to 2024. *Atmos. Chem. Phys.*, **2026** 26, 3783–3803, <https://doi.org/10.5194/acp-26-3783-2026>.
- 2026 **Nawaz, M. O.**; Henze, D. K. National Climate Action Can Ameliorate, Perpetuate, or Exacerbate International Air Pollution Inequalities. *Nature Communications* **2026**. <https://doi.org/10.1038/s41467-026-68827-0>.
- 2025 Siu, T. K.; Goldberg, D. L.; Kerr, G. H.; Chen, L.; **Nawaz, M. O.**; Chang, R. Y.-W.; Fong, K. C. Tropospheric NO₂ Patterns in Eastern Canada Using the First Year of TEMPO Observations. *Journal of Geophysical Research: Atmospheres* **2025**, 130 (24), e2025JD044757. <https://doi.org/10.1029/2025JD044757>.
- 2025 Goldberg, D. L.; **Nawaz, M. O.**; Lyu, C.; He, J.; Carlton, A. G.; Kondragunta, S.; Anenberg, S. C. Clear-Sky and Cloudy-Sky Differences in NO₂ Concentrations over the United States: Implications for Satellite Measurement Applications. *Atmospheric Chemistry and Physics* **2025**, 25 (22), 16287–16302. <https://doi.org/10.5194/acp-25-16287-2025>.
- 2025 **Nawaz, M. O.**; Huber, D. E.; Kerr, G. H.; Judd, L. M.; Acker, S. J.; Goldberg, D. L. A Comparative Analysis of TEMPO NO₂ Remote Sensing With Surface-Level

Monitoring Through Diurnal and Seasonal Trends, Meteorology, and Monitor Characteristics. *Journal of Geophysical Research: Atmospheres* **2025**, 130 (20), e2025JD043923. <https://doi.org/10.1029/2025JD043923>.

- 2025 Hay, S. I.; Ong, K. L.; Santomauro, D. F.; ... **Nawaz, M. O.**; ... et al. Burden of 375 Diseases and Injuries, Risk-Attributable Burden of 88 Risk Factors, and Healthy Life Expectancy in 204 Countries and Territories, Including 660 Subnational Locations, 1990–2023: A Systematic Analysis for the Global Burden of Disease Study 2023. *The Lancet* **2025**, 406 (10513), 1873–1922. [https://doi.org/10.1016/S0140-6736\(25\)01637-X](https://doi.org/10.1016/S0140-6736(25)01637-X).
- 2025 Jin, L.; Benoit, J.; **Nawaz, M. O.**; Rodrigues, P. F.; Wiecko, P.; Miller, J.; Alvarez, G.; Henze, D. K.; Osipova, L.; Anenberg, S. C. Global Health Benefits of Policies to Reduce On-Road Vehicle Pollution through 2040. *Environ. Res. Lett.* **2025**. <https://doi.org/10.1088/1748-9326/adcd87>.
- 2025 Wiecko, P.; Henze, D. K.; **Nawaz, M. O.** Sector-, Season-, and Country-Specific NO₂-Associated Health Benefits from NO_x Emission Reductions. *ACS EST Air* **2025**, 2 (4), 700–709. <https://doi.org/10.1021/acsestair.5c00012>.
- 2025 **Nawaz, M. O.**; Goldberg, D. L.; Kerr, G. H.; Anenberg, S. C. TROPOMI Satellite Data Reshape NO₂ Air Pollution Land-Use Regression Modeling Capabilities in the United States. *ACS EST Air* **2025**, 2 (2), 187–200. <https://doi.org/10.1021/acsestair.4c00153>.
- 2024 Dyer, G. M. C.; Khomenko, S.; Adlakha, D.; Anenberg, S.; Angelova, J.; Behnisch, M.; Boeing, G.; Chen, X.; Cirach, M.; de Hoogh, K.; Diez Roux, A. V.; Esperon-Rodriguez, M.; Flueckiger, B.; Gasparrini, A.; Iungman, T.; Khreis, H.; Kondo, M. C.; Masselot, P.; McDonald, R. I.; Montana, F.; Mitchell, R.; Mueller, N.; **Nawaz, M. O.**; Pereira, E.; Pisoni, E.; Prieto-Curiel, R.; Rezaei, N.; Rybski, D.; Ramasco, J. J.; Schifanella, R.; Shabou, S.; Tatah, L.; Taubenböck, H.; Tonne, C.; Velázquez-Cortés, D.; Woodcock, J.; Zhang, Q.; Nieuwenhuijsen, M. Commentary: A Road Map for Future Data-Driven Urban Planning and Environmental Health Research. *Cities* **2024**, 155, 105340. <https://doi.org/10.1016/j.cities.2024.105340>.
- 2024 Goldberg, D. L.; de Foy, B.; **Nawaz, M. O.**; Johnson, J.; Yarwood, G.; Judd, L. Quantifying NO_x Emission Sources in Houston, Texas Using Remote Sensing Aircraft Measurements and Source Apportionment Regression Models. *ACS EST Air* **2024**, 1 (11), 1391–1401. <https://doi.org/10.1021/acsestair.4c00097>.
- 2024 Dyer, G. M. C.; Khomenko, S.; Adlakha, D.; Anenberg, S.; Behnisch, M.; Boeing, G.; Esperon-Rodriguez, M.; Gasparrini, A.; Khreis, H.; Kondo, M. C.; Masselot, P.; McDonald, R. I.; Montana, F.; Mitchell, R.; Mueller, N.; **Nawaz, M. O.**; Pisoni, E.; Prieto-Curiel, R.; Rezaei, N.; Taubenböck, H.; Tonne, C.; Velázquez-Cortés, D.; Nieuwenhuijsen, M. Exploring the Nexus of Urban Form, Transport, Environment and Health in Large-Scale Urban Studies: A State-of-the-Art Scoping Review. *Environmental Research* **2024**, 257, 119324. <https://doi.org/10.1016/j.envres.2024.119324>.
- 2024 Choi, J.; Henze, D. K.; **Nawaz, M. O.**; Malley, C. S. Source Attribution of Health Burdens From Ambient PM_{2.5}, O₃, and NO₂ Exposure for Assessment of South

Korean National Emission Control Scenarios by 2050. *GeoHealth* **2024**, 8 (8), e2024GH001042. <https://doi.org/10.1029/2024GH001042>.

- 2024 **Nawaz, M. O.**; Johnson, J.; Yarwood, G.; de Foy, B.; Judd, L.; Goldberg, D. L. An Intercomparison of Satellite, Airborne, and Ground-Level Observations with WRF–CAMx Simulations of NO₂ Columns over Houston, Texas, during the September 2021 TRACER-AQ Campaign. *Atmospheric Chemistry and Physics* **2024**, 24 (11), 6719–6741. <https://doi.org/10.5194/acp-24-6719-2024>.
- 2023 Gu, Y.; Henze, D. K.; **Nawaz, M. O.**; Wagner, U. J. Response of the Ozone-Related Health Burden in Europe to Changes in Local Anthropogenic Emissions of Ozone Precursors. *Environ. Res. Lett.* **2023**, 18 (11), 114034. <https://doi.org/10.1088/1748-9326/ad0167>.
- 2023 **Nawaz, M. O.**; Henze, D. K.; Huneus, N. J.; Osses, M.; Álamos, N.; Opazo, M. A.; Gallardo, L. Sources of Air Pollution Health Impacts and Co-Benefits of Carbon Neutrality in Santiago, Chile. *Journal of Geophysical Research: Atmospheres* **2023**, 128 (19), e2023JD038808. <https://doi.org/10.1029/2023JD038808>.
- 2023 Jo, D. S.; Nault, B. A.; Tilmes, S.; Gettelman, A.; McCluskey, C. S.; Hodzic, A.; Henze, D. K.; **Nawaz, M. O.**; Fung, K. M.; Jimenez, J. L. Global Health and Climate Effects of Organic Aerosols from Different Sources. *Environ. Sci. Technol.* **2023**, 57 (37), 13793–13807. <https://doi.org/10.1021/acs.est.3c02823>.
- 2023 Gu, Y.; Henze, D. K.; **Nawaz, M. O.**; Cao, H.; Wagner, U. J. Sources of PM_{2.5}-Associated Health Risks in Europe and Corresponding Emission-Induced Changes During 2005–2015. *GeoHealth* **2023**, 7 (3), e2022GH000767. <https://doi.org/10.1029/2022GH000767>.
- 2023 **Nawaz, M. O.**; Henze, D. K.; Anenberg, S. C.; Ahn, D. Y.; Goldberg, D. L.; Tessum, C. W.; Chafe, Z. A. Sources of Air Pollution-Related Health Impacts and Benefits of Radially Applied Transportation Policies in 14 US Cities. *Front. Sustain. Cities* **2023**, 5. <https://doi.org/10.3389/frsc.2023.1102493>.
- 2023 **Nawaz, M. O.**; Henze, D. K.; Anenberg, S. C.; Braun, C.; Miller, J.; Pronk, E. A Source Apportionment and Emission Scenario Assessment of PM_{2.5}- and O₃-Related Health Impacts in G20 Countries. *GeoHealth* **2023**, 7 (1), e2022GH000713. <https://doi.org/10.1029/2022GH000713>.
- 2021 **Nawaz, M. O.**; Henze, D. K.; Harkins, C.; Cao, H.; Nault, B.; Jo, D.; Jimenez, J.; Anenberg, S. C.; Goldberg, D. L.; Qu, Z. Impacts of Sectoral, Regional, Species, and Day-Specific Emissions on Air Pollution and Public Health in Washington, DC. *Elementa: Science of the Anthropocene* **2021**, 9 (1), 00043. <https://doi.org/10.1525/elementa.2021.00043>.
- 2021 Cao, H.; Henze, D. K.; Cady-Pereira, K.; McDonald, B. C.; Harkins, C.; Sun, K.; Bowman, K. W.; Fu, T.-M.; **Nawaz, M. O.** COVID-19 Lockdowns Afford the First Satellite-Based Confirmation That Vehicles Are an Under-Recognized Source of Urban NH₃ Pollution in Los Angeles. *Environ. Sci. Technol. Lett.* **2021**, 9 (1), 3–9. <https://doi.org/10.1021/acs.estlett.1c00730>.
- 2021 Nault, B. A.; Jo, D. S.; McDonald, B. C.; Campuzano-Jost, P.; Day, D. A.; Hu, W.; Schroder, J. C.; Allan, J.; Blake, D. R.; Canagaratna, M. R.; Coe, H.; Coggon, M. M.;

DeCarlo, P. F.; Diskin, G. S.; Dunmore, R.; Flocke, F.; Fried, A.; Gilman, J. B.; Gkatzelis, G.; Hamilton, J. F.; Hanisco, T. F.; Hayes, P. L.; Henze, D. K.; Hodzic, A.; Hopkins, J.; Hu, M.; Huey, L. G.; Jobson, B. T.; Kuster, W. C.; Lewis, A.; Li, M.; Liao, J.; **Nawaz, M. O.**; Pollack, I. B.; Peischl, J.; Rappenglück, B.; Reeves, C. E.; Richter, D.; Roberts, J. M.; Ryerson, T. B.; Shao, M.; Sommers, J. M.; Walega, J.; Warneke, C.; Weibring, P.; Wolfe, G. M.; Young, D. E.; Yuan, B.; Zhang, Q.; de Gouw, J. A.; Jimenez, J. L. Secondary Organic Aerosols from Anthropogenic Volatile Organic Compounds Contribute Substantially to Air Pollution Mortality. *Atmospheric Chemistry and Physics* **2021**, *21* (14), 11201–11224. <https://doi.org/10.5194/acp-21-11201-2021>.

2021 Malley, C. S.; Hicks, W. K.; Kulyenstierna, J. C. I.; Michalopoulou, E.; Molotoks, A.; Slater, J.; Heaps, C. G.; Ulloa, S.; Veysey, J.; Shindell, D. T.; Henze, D. K.; **Nawaz, M. O.**; Anenberg, S. C.; Mantlana, B.; Robinson, T. P. Integrated Assessment of Global Climate, Air Pollution, and Dietary, Malnutrition and Obesity Health Impacts of Food Production and Consumption between 2014 and 2018. *Environ. Res. Commun.* **2021**, *3* (7), 075001. <https://doi.org/10.1088/2515-7620/ac0af9>.

2020 **Nawaz, M. O.**; Henze, D. K. Premature Deaths in Brazil Associated With Long-Term Exposure to PM_{2.5} From Amazon Fires Between 2016 and 2019. *GeoHealth* **2020**, *4* (8), e2020GH000268. <https://doi.org/10.1029/2020GH000268>.

Manuscripts Under Review (7)

2026 Gu, Y, Henze, D.K., **Nawaz, M.O.**, Zhang, L., Liao, H. Non-agricultural NH₃ emissions disproportionately contribute to urban PM_{2.5} health burdens in China. *Under Review*.

2026 Graffy, P.M., ...**Nawaz, M.O.**, ... et al. Respiratory Trajectory Phenotypes and Air Pollution-Associated Outcomes in Critically Ill Patients with Lung Cancer in the United States. *Under Review*.

2026 Goldberg, D.L., Johnson, J., **Nawaz, M.O.**, Kerr, G.H., Chen, L., Huber, D.E., Judd, L. M. Contrasting the TEMPO and TROPOMI Views of Tropospheric NO₂: Implications for Diurnal Variability and Model Evaluation. *Under Review*.

2026 Graffy, P.M., ...**Nawaz, M.O.**, ... et al. Cumulative Air Pollution Exposure Increases the Severity of Acute Respiratory Failure: a U.S. Multicenter Study. *Under Review*

2025 Morris, S. T., O'Neill, B. C., Msangi, S., **Nawaz, M.O.**, Parker, N., Rao, N., Van Vuuren, D., et al. Modeling the human well-being dimensions of global change: priorities and challenges for research to inform decision-making. *Under Review*

2025 TOAR-II Health Collaborators, Tropospheric Ozone Assessment Report, Phase II: Changing Health Impacts of Global Ozone Exposure. *Under Review*

2025 Kerr, G.H., **Nawaz, M.O.**, Anenberg, S.C., Anthoff, D., Burton, C., Carter, T.S., Henze, D.K., Kelley, D.I., Kingdon, C., O'Dell, K., Prest, B.C., Cromar, K.R. Climate-driven surges in public health damages from wildland fire-sourced pollution through 2100. *Under Review*

Manuscripts In Preparation (2)

TBD	Kim, S.Y., Kerr, G.H., Nawaz, M.O. , Anenberg, S.C. Fine-scale spatiotemporal patterns of NO ₂ pollution and associated mortality burdens across the continental United States. <i>In preparation</i> .
TBD	Nawaz, M.O. and Henze, D.K. Public health damages from reversal of declining coal-fired electricity generation in the United States. <i>In preparation</i> .

HONORS AND FELLOWSHIPS

2026	Cardiff University Enriching Student Life Awards Nominee
2026	GW4 Crucible 2026 Cohort
2024-2025	GeoCAFE Scholar
2023-2024	Natural Resources Defense Council Health Science Policy Fellowship
2018	Outstanding Mechanical Engineering Research Potential Fellowship
2018	1 st place student poster award, UNC 5 th Climate Change Symposium

RESEARCH SUPERVISION

Current

2026-	Lucas Ngo (Postdoc)
2025-	Feichi Hu (PhD) (Co-supervised with L. Zhuo)
2025-	Kyla Camama (MSci)

Past

2025-2026	Niamh Delamar (BSc)
2025-2026	Alice Gittoes-Davies (BSc)
2025-2026	Carwyn Jones (BSc)
2025-2026	Olivia Pease (BSc)
2024-2025	Olivia Paquette (BSc)
2023-2024	Katie O'Donnell (MPH)
2023-2024	Erin Campbell (MPH)
2019-2021	Mohammed Alwakeel (BSc)

PhD Examination

2026	Patrick Wiecko; panel member for comprehensive exam
2025	Feichi Hu; panel member for 9-month review

Informal Supervision

2025	Pastoral mentor for the EQUATOR Project for equity in environmental research
2023-	Advisor for PhD student on transportation pollution modeling project for the ICCT
2024	Advisor for PhD student on accepted NASA FINESST proposal

TEACHING

Cardiff University, School of Earth and Environmental Sciences

2026	Climate Change Adaptation and Resilience
Spring	Lecturer for Postgraduate Module of ~20 Students
	Digital Fieldwork Workshop
	Lecturer for Year 1 Coding Workshop for ~120 Students
	MSci Environmental Geoscience Dissertations

Supervisor and/or Marker for 2 MSci Dissertations

2025 **The Ocean-Atmosphere System**
Fall Module Leader for Year 2 Undergraduate Module of ~130 Students

GIS, Maps, and Analytical Skills
Lecturer for Year 1 Undergraduate Module of ~220 Students

BSci Environmental Geography Dissertations
Supervisor and/or Marker for ~20 Year 3 Undergraduate Dissertations

Environmental and Physical Geography Students
Tutor for Year 1/2/3 Undergraduate Tutorials

2025 **Digital Fieldwork Workshop**
Spring Lecturer for Year 1 Undergraduate Workshop of ~130 Students

George Washington University, Department of Environmental and Occupational Health

2024 **Global Climate Change & Air Pollution**
Fall Professorial Lecturer for MPH Course of ~5 Students

University of Colorado Boulder, Department of Mechanical Engineering

2020 **Fluid Mechanics and Thermodynamics Exams**
Summer Student Lead and Mentor for Review Course of ~20 PhD Students

2019 **Numerical Methods for Engineers**
Spring Lead Teaching Assistant for Undergraduate Course of ~100 Students

2018 **Numerical Methods for Engineers**
Fall Teaching Assistant for Undergraduate Course of ~100 Students

PROFESSIONAL ASSOCIATIONS

Current

2025- TOAR-II Health Team
2025- COST Action CliMent Member (CA23113)
2024- Global Burden of Disease Study Collaborator
2023- European Geophysical Union
2018- American Geophysical Union

Past

2025 EQUATOR Mentor
2024-2025 American Chemistry Society (ACS)
2023-2024 American Meteorological Society

MEDIA COVERAGE

BBC News, *How air pollution has changed since traditional steelmaking ended in Port Talbot*, <https://www.bbc.co.uk/news/articles/cjdr0zp273vo>, 23 December 2025

BBC News Radio Wales, *Cleaner air in steel town a year since furnace shut*, <https://www.bbc.co.uk/sounds/play/p0mq8hsb>, 23 December 2025

WXPR, *The health impacts of tailpipe emissions get a fresh look*, <https://www.wxpr.org/energy-environment/2025-08-11/the-health-impacts-of-tailpipe-emissions-get-a-fresh-look>, 11 August 2025

- Scientific American, *How Air Pollution Damages the Brain*,
<https://www.scientificamerican.com/article/what-is-pollution-doing-to-our-brains-exposomics-reveals-links-to-many/>, 12 April 2024
- EOS Earth and Space Science News, *The Global Health Benefits of Going Net Zero*,
<https://eos.org/research-spotlights/the-global-health-benefits-of-going-net-zero>, 14 February 2023
- U.S. News, *Pandemic Data Suggests Cars Spew More Ammonia Than Suspected*,
<https://www.usnews.com/news/health-news/articles/2021-12-03/pandemic-data-suggests-cars-spew-more-ammonia-than-suspected>, 3 December 2021
- Revista Pesquisa Tapesp, *Queimadas na Amazônia aumentam internações* [*Fires in the Amazon increase hospitalizations*], <https://revistapesquisa.fapesp.br/queimadas-na-amazonia-aumentam-internacoes/>, 18 September 2020

DEPARTMENTAL SERVICE

Cardiff University, School of Earth and Environmental Science

- 2026 Member of School Promotions Panel for Earth and Environmental Science
- 2026- Member of Extenuating Circumstances Panel
- 2025- Director of Equality, Diversity, and Inclusion
- 2025- Member of Ethics Committee
- 2025 Member of School Research Leave Application Review Panel
- 2025 Organiser for EDI Away Day
- 2025 Induction Speaker for Year 1 Students School of Earth and Environmental Science
- 2025 Volunteer for Open Day for School of Earth and Environmental Science

ACADEMIC SERVICE

Ad-Hoc Peer-Review for Journals (40)

- 2026 *Communications Earth & Environment* (1); *Earth System Science Data* (1); *Geoscience Letters* (1); *Nature Health* (1); *Nature Sustainability* (1)
- 2025 *Discover Atmospheres* (1); *Environmental Monitoring & Assessment* (1); *Environmental Research Letters* (1); *Environmental Science Policy Research* (1); *Humanities & Social Science Communications* (1); *Nature Cities* (1); *Nature Communications* (1); *Nature Health* (3); *NPJ Clean Air* (1); *NPJ Climate & Atmospheric Science* (1); *PLOS* (1)
- 2024 *Atmospheric Chemistry and Physics* (1); *Discover Cities* (1); *Environmental Research Letters* (1); *Environmental Science & Technology Air* (1); *GeoHealth* (2); *Health Data Science* (1); *International Geoscience and Remote Sensing Symposium* (6); *Nature Food* (1); *Scientific Reports* (1)
- 2023 *Environmental Science & Technology* (1); *GeoHealth* (1)
- 2022 *Environmental Science & Technology* (1)
- 2021 *Elementa: Science of the Anthropocene* (1); *Lancet Planetary Health* (1)
- 2019 *Environmental Science & Technology* (1); *Journal of the Air and Waste Management Association* (1)

Ad-Hoc Peer-Review for Proposals (2)

- 2025 Wellcome Trust Expert Reviewer (2)

Editorial Roles

- 2026- Associate Editor for *Air Quality, Atmosphere & Health*

2025 Associate Editor for *Frontiers in Environmental Health* (Environmental Epidemiology)

Other Professional Service

- 2026 Session convener on air quality, climate, health, and equity for EGU 2026 meeting
2025 Session convener on air quality, climate, health, and equity for EGU 2025 meeting
2017 Conference coordinator for the CMAS 2017 meeting

PRESENTATIONS (31)

Invited Talks (4)

- 2026 The added benefit of TEMPO remote sensing of NO₂ for exposure and health impact assessment. June 1st-5th. CEOS Atmospheric Composition-Virtual Constellation. Leiden, Netherlands.
- 2026 Using HPC for the variable resolution simulation of air pollution formation using the GCHP stretched-grid model. May 20th. Advanced Research Computing Operational & User Group meeting. Cardiff, Wales, UK.
- 2025 Applying machine learning and statistical modeling approaches to remote sensing observations for inferring surface-level NO₂. December 15th-19th. American Geophysical Union. New Orleans, LA, USA.
- 2023 Using satellite data to characterize air pollution and health in cities and countries. April 27th. Environmental Defense Fund / Climate and Clean Air Coalition / United Nations Environmental Programme Workshop for Clean Air Solutions in Latin America and the Caribbean. Bogotá, Colombia.

Oral Presentations (12)

- 2025 **Nawaz, M.O.**, Henze, D.K., Exploring the role of climate action in transboundary air pollution inequality using GEOS-Chem adjoint sensitivities. April 29th. European Geophysical Union. Vienna, Austria
- 2024 **Nawaz, M.O.**, Anenberg, S.C., Goldberg, D.L., Kerr, G.H., Kondragunta, S. Development of a Land-Use Regression of Hourly Surface NO₂ in preparation for GeoXO Atmospheric Composition Data. April 17th. European Geophysical Union. Vienna, Austria.
- 2024 **Nawaz, M.O.**, O'Dell, K., Anenberg, S.C., Goldberg, D.L., Kerr, G.H., He, J., McDonald, B., Kondragunta, S. Value of GeoXO Atmospheric Composition Data for Estimating Air Pollution-Related Health Impacts. January 30th. American Meteorological Society. Baltimore, MD, USA.
- 2023 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Goldberg, D.L., Investigating climate co-benefits using GEOS-Chem adjoint sensitivities. August 15th. Second GEOS-Chem Europe Meeting. London, UK.
- 2023 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Tessum, C. Regional vs local sources of municipal air pollution-related health impacts. January 10th. American Meteorological Society. (Presented by Henze). Denver, CO, USA.
- 2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Huang, T. Developing an interactive tool for characterizing the air pollution-related health impacts in Los Angeles, CA associated

- with different proposed emission scenarios. July 19th. Earth Science Information Partners Meeting. Pittsburgh, PA, USA (Virtual).
- 2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Harkins, C., Gallardo, L., Barazza Basoa, K. Leveraging satellite-derived data in GEOS-Chem adjoint simulations to characterize the sources of PM_{2.5}-, O₃-, and NO₂-related health impacts at multiple spatial scales. June 9th. 10th International GEOS-Chem Meeting. St. Louis, MO, USA. (Virtual).
- 2022 **Nawaz, M.O.**, Henze, D.K., Braun, C., Miller, J., Pronk, E., Anenberg, S.C. Characterizing the sources of air pollution at the urban- and country-scale: case studies in Santiago, Chile and G20 countries. February 17th. Graduate Engineering Annual Research and Recruitment Symposium. Boulder, CO, USA.
- 2021 **Nawaz, M.O.**, D. Henze, S.C. Anenberg, C. Braun, J. Miller. Comparing domestic and extra-regional contributions to pollutant exposures and health impacts in G20 countries through a novel adjoint modeling approach. December 15th. American Geophysical Union Fall Meeting. New Orleans, LA, USA (Virtual).
- 2020 **Nawaz, M.O.**, D. Henze, D. Goldberg, S. Anenberg, D. Jo, B. Nault, J.L. Jimenez, H. Cao, C. Harkins, Z. Qu. Characterizing the regional, sectoral and species-specific sources of pollution exposure and its associated health impacts in urban environments: case studies in Washington, D.C. and Santiago, Chile. December 14th. American Geophysical Union. (Virtual)
- 2020 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Goldberg, D. Premature deaths in Brazil associated with long-term exposure to PM_{2.5} from Amazon fires and development of a nested South American domain for the GEOS-Chem Adjoint. June 23rd. 19th GEIA Conference. (Virtual)
- 2019 **Nawaz, M.O.**, Henze, D.K. Source attribution of PM_{2.5} from sensitivity analyses in the GEOS-Chem adjoint model. October 25th. Young Scientists Symposium on Atmospheric Research. Fort Collins, CO, USA.

Poster Presentations (15)

- 2026 **Nawaz, M.O.**, Goldberg, D.G., Anenberg, S.C., Kerr, G.H. Estimating the air pollutant-attributable health burden of the oil and gas sector in Mexico using a TROPOMI flux-divergence approach. May 7th. European Geosciences Union. Vienna, Austria.
- 2025 **Nawaz, M.O.**, Goldberg, D.G., Anenberg, S.C., Kerr, G.H. What does low-earth orbiting, geostationary, and airborne remote-sensing reveal about surface NO₂? September 9th. UK Atmospheric Chemistry Conference. York, UK.
- 2025 **Nawaz, M.O.**, Southerland, V.A., Goldberg, D.G. Characterizing the air quality and health impacts from oil and gas emissions in Mexico using GCHP. August 19th. GEOS-Chem Europe Meeting 3. London, UK.
- 2024 **Nawaz, M.O.**, Goldberg, D.L., Kerr, G.H., Anenberg, S.C., What can TROPOMI and TEMPO remote sensing reveal about seasonal and diurnal trends in surface-level NO₂? December 13th. American Geophysical Union. Washington, DC, USA.
- 2022 **Nawaz, M.O.**, Henze, D.K., Anenberg, S.C., Harkins, C., Gallardo, L., Barazza Basoa, K. Leveraging satellite-derived data and air quality modeling to characterize source profiles

- of climate co-benefits at the urban- and country- scale. December 12th. American Geophysical Union. Chicago, IL, USA.
- 2020 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, A. Van Donkelaar, R. Martin, M. L. Serre, J. J. West. Health benefits of decreases in PM_{2.5} and ozone in the United States, 1990-2016. July 21st. NASA Health and Air Quality Applied Sciences Team Final Showcase. (Virtual).
- 2019 **Nawaz, M.O.**, D.K. Henze, S.C. Anenberg, D. Goldberg, Z. Qu (2019). Source attribution of PM_{2.5} and O₃ concentrations and health outcomes from 2010 and 2011 in Washington D.C. using sensitivity analyses in the GEOS-Chem adjoint model. December 19th. American Geophysical Union, San Francisco, CA, USA
- 2019 **Nawaz, M.O.**, D.K. Henze, C.S. Malley, J.C.I. Kuylenstierna, H.W. Vallack, Y. Davila, S.C. Anenberg, S. Terry, A. Curry-Brown, N. Fann, E. Lefevre, C. Heaps, S. Penn, H. Roman, J. Neumann. Source attribution of climate and health impacts from aerosols. May 6th. 9th International GEOS-Chem Meeting, Cambridge, MA, USA.
- 2019 **Nawaz, M.O.**, Henze, D.K., The use of adjoint modeling to assess the sources of air pollution and its associated health impacts. February 21st. Graduate Engineering Annual Research and Recruitment Symposium. Boulder, CO, USA.
- 2018 **Nawaz, M.O.**, D. K. Henze, C. Malley, GH41C-1446: Source Attribution of Climate and Health Impacts from Aerosols. February 14th. AGU Fall Meeting. Washington, DC, USA.
- 2018 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, A. van Donkelaar, R. V. Martin, J. J. West. Health benefits of decreases in PM_{2.5} and ozone in the United States, 1990-2015. July 16th. NASA Health and Air Quality Applied Sciences Team Meeting. Madison, WI, USA.
- 2018 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, A. van Donkelaar, R. V. Martin, J. J. West (2018). Health benefits of decreases in PM_{2.5} and ozone in the United States, 1990-2015. April 20th. Climate Change and Resilience Symposium. Chapel Hill, NC, USA.
- 2017 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, J. J. West. Health benefits of decreases in PM_{2.5} and ozone in the United States from 1990 to 2015. December 11th. American Geophysical Union. New Orleans, LA, USA.
- 2017 **Nawaz, M.O.**, Y. Zhang, D. Q. Tong, J. J. West. Health benefits of decreases in PM_{2.5} and ozone in the United States from 1990 to 2015. October 23rd. Community Modeling and Analysis System Conference. Chapel Hill, NC, USA.
- 2017 **Nawaz M.O.**, Y. Zhang, West, J.J. Impact of regional ozone precursor emissions on global ozone burden. April 12th. Celebration of Undergraduate Research. Chapel Hill, NC, USA.